NewsBot Intelligence System

ITAI 2373 | Midterm Technical Report

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Group: Individual

GitHub repository: https://github.com/zainahm713/ITAI2373-Portfolio

Data source: BBC News labeled dataset, referenced at https://www.kaggle.com/competitions/learn-ai-bbc/data

1. Executive summary and business case

This project implements an end-to-end news intelligence pipeline on 1440 usable labeled English articles in 5 categories. Potential users include media analysts, editorial teams, and corporate communications staff. The system combines categorization, sentiment cues, named-entity discovery, and linguistic descriptions to reduce manual triage of news articles. The output is decision support, not a verified account of causation, human emotion, or economic outcomes.

2. Data preparation

Loaded **BBC News Train.csv**, removed missing, short, and duplicate text, and applied stratified sampling if needed. Final data: **1440** articles. Labels and counts: sport: 342; business: 335; politics: 266; entertainment: 263; tech: 234.

The preprocessing pipeline performs whitespace/URL normalization, spaCy tokenization, stop-word and nonalphabetic-token removal, and lemmatization. For spaCy feature extraction, each article was capped at 5000 characters, a documented resource limitation.

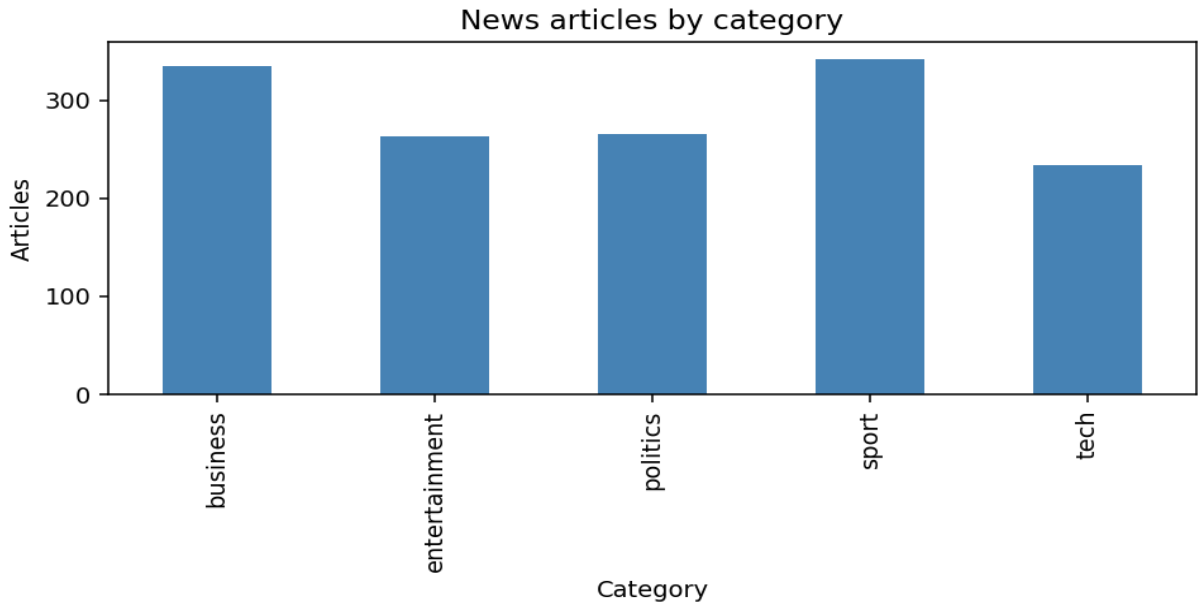


Figure 1. Dataset category distribution.

3. TF-IDF: interpretable lexical features

TF-IDF uses unigrams and bigrams, a maximum of 7,000 vocabulary features, document-frequency bounds, and sublinear term frequencies. To avoid leakage, the vocabulary was fitted to the training split only.

business: say, firm, company, market, year

entertainment: film, star, award, actor, say

politics: labour, election, government, say, blair

sport: win, game, play, match, player

tech: people, user, technology, say, mobile

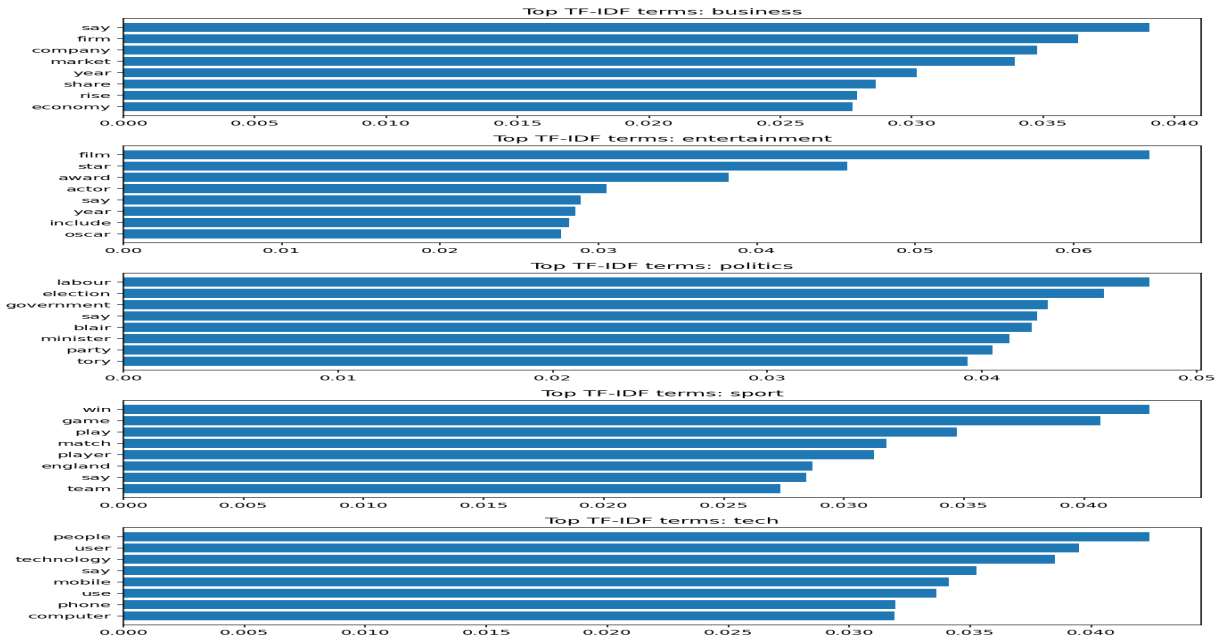


Figure 2. Top class-level training-set TF-IDF terms.

A lecture-based Bag-of-Words and Naive Bayes baseline yielded 96.4% accuracy and 0.962 macro F1 on held-out articles. Training-only chi-square analysis identified terms associated with each label.

Chi-square (business): bank, economy, growth

Chi-square (entertainment): film, star, actor

Chi-square (politics): labour, tory, blair

Chi-square (sport): match, coach, cup

Chi-square (tech): user, software, technology

4. Multi-class classification: model selection and evaluation

The dataset was split into stratified training (1080) and held-out test (360) examples. Multinomial Naive Bayes, Logistic Regression and Linear SVM were evaluated. The model was chosen by three-fold **training-only** macro-F1 cross-validation, not by looking at the holdout performance.

Model	CV macro F1	Test accuracy	Test macro F1
LinearSVC	0.974	0.964	0.961
LogisticRegression	0.966	0.961	0.959

Model	CV macro F1	Test accuracy	Test macro F1
MultinomialNB	0.966	0.958	0.955

Training-CV selected model: **LinearSVC**; held-out accuracy 96.4%, and held-out macro F1 0.961. These numbers describe this split, not guaranteed performance on new events.

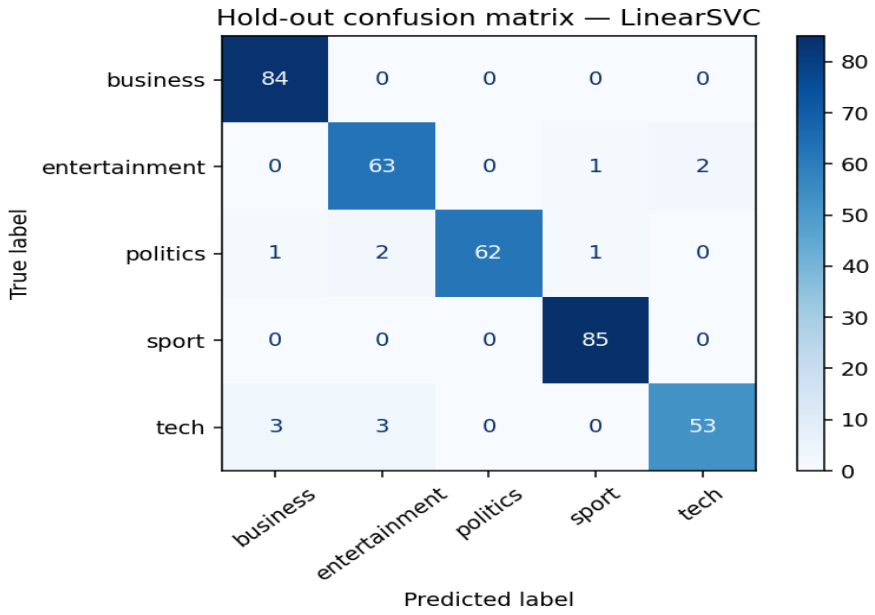


Figure 3. Confusion matrix on previously held-out articles.

Most frequent confusion observed: 3 'tech' test articles were predicted as 'entertainment'.

5. POS, syntax, and style analysis

Computationally expensive syntax/POS/NER processing was performed on a stratified sample of **175** articles (up to 35 per category). Normalized POS frequencies, average sentence length, adjective–noun patterns, and approximations of grammatical subject–verb–object structures were extracted.

business: noun share 25.2%, verb share 12.3%, proper-noun share 7.6%.

entertainment: noun share 25.5%, verb share 12.0%, proper-noun share 10.5%.

politics: noun share 23.1%, verb share 14.2%, proper-noun share 7.6%.

sport: noun share 19.6%, verb share 12.9%, proper-noun share 11.2%.

tech: noun share 26.1%, verb share 13.3%, proper-noun share 4.7%.

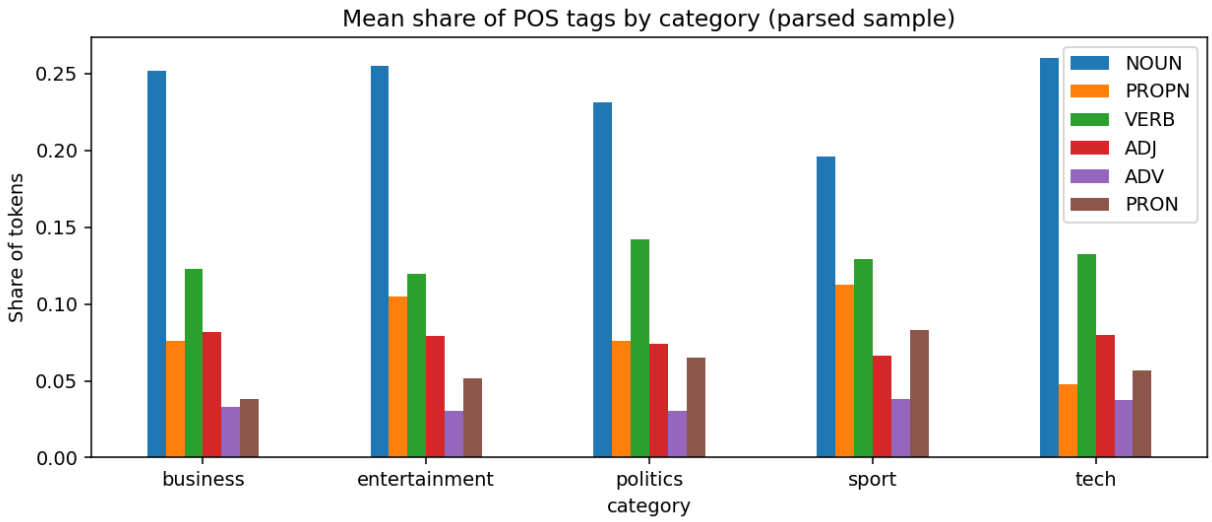


Figure 4. Relative POS tag frequencies by category in sampled documents.

6. Sentiment and emotion cues

NLTK VADER scored each article with a lexical compound sentiment value. A small exploratory dictionary counted words related to fear, anger, joy, and sadness. These word cues are not validated emotion predictions and should not be interpreted as author intent.

business: negative 36.4%, neutral 1.5%, positive 62.1%

entertainment: negative 19.0%, neutral 0.8%, positive 80.2%

politics: negative 46.6%, neutral 0.4%, positive 53.0%

sport: negative 20.8%, neutral 0.3%, positive 78.9%

tech: negative 20.9%, neutral 1.3%, positive 77.8%

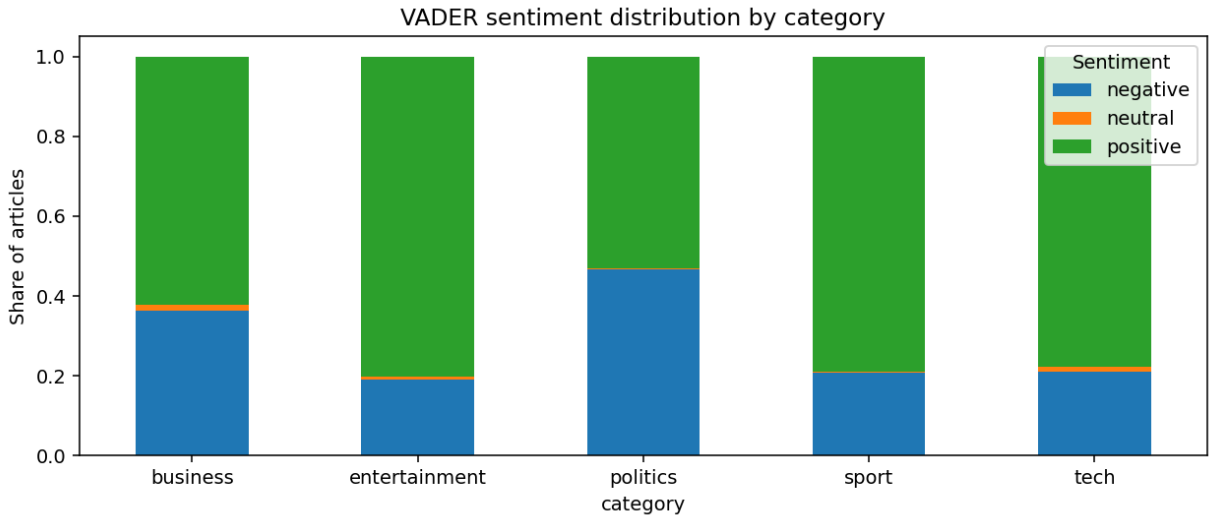


Figure 5. Fraction of articles assigned each VADER sentiment label.

7. Named entities and co-mentions

The en_core_web_sm recognizer extracted PERSON, ORG, GPE, DATE and MONEY mentions. Frequencies were compared by topic; repeated co-mentions describe article overlap and do not establish real-world business relationships.

Frequently detected mentions: uk (GPE: 57); us (GPE: 56); london (GPE: 42); 2004 (DATE: 41); britain (GPE: 37); last year (DATE: 29); sunday (DATE: 29); bbc (ORG: 26).

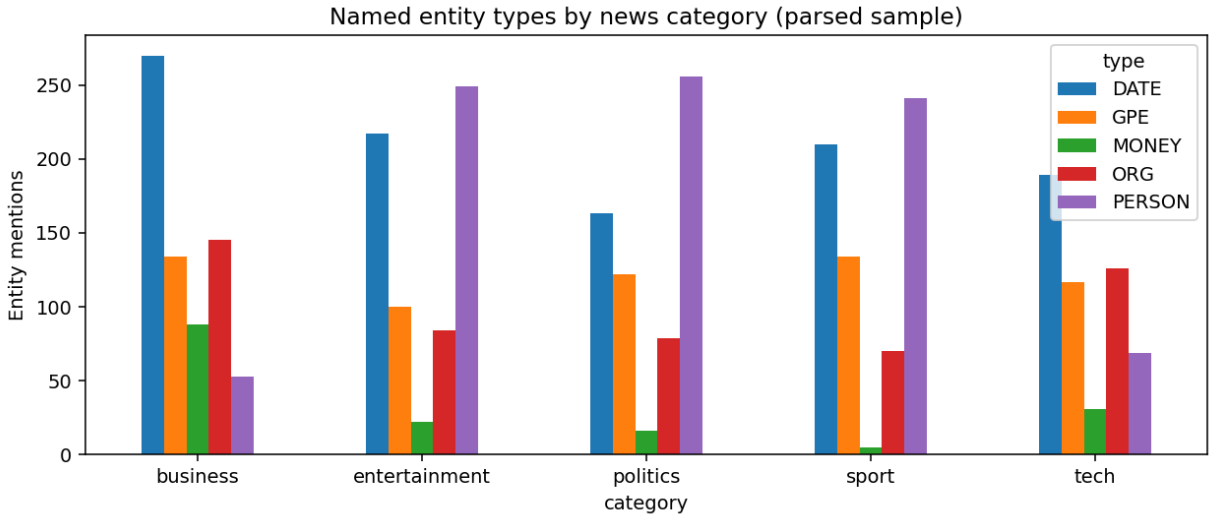


Figure 6. NER label frequencies across sampled articles.

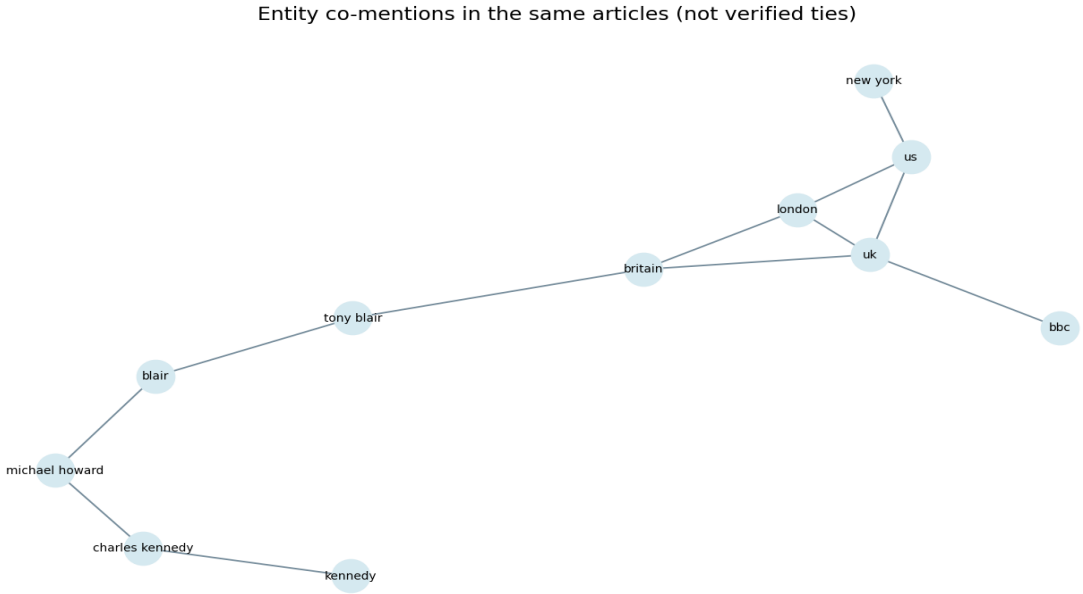


Figure 7. Entity co-mentions. Edges do not establish real-world ties.

8. End-to-end pipeline demonstration

The combined function accepts article text, cleans and lemmatizes it, predicts a news category, assigns a VADER sentiment score, and returns detected entities and common POS tags. Two illustrative **smoke-test** passages check system integration; they are not independent real-world validation.

Technology example: predicted tech; VADER positive (+0.700); recognized date: Monday; extracted relation: company - announce - tool.

Sport example: predicted sport; VADER positive (+0.955); recognized date: Saturday; extracted relation: team - win - tournament.

9. Business implications, critical evaluation, limitations

Observed patterns: 3 'tech' test articles were predicted as 'entertainment'. politics had the highest VADER-negative share (46.6%); sport had the highest proper-noun share in sampled texts (11.2%). These descriptive observations should not be interpreted as evidence of cause or economic value.

Potential applications include topic routing for editors, high-volume coverage monitoring for public relations teams, and exploratory trend summaries for business analysts. The saved CSVs and figures support human investigation but do not directly prove commercial impact. Observed category words may reflect the historical BBC dataset rather than current reporting.

Risks include class imbalance, historical domain shift, name ambiguities, misleading word-based sentiment, parsing errors in passive clauses, and false inferences from entity co-occurrence. A future evaluation should use fresh articles with human annotations, check subgroup and genre performance, and document false positives.

10. Implementation and reproducibility

All reported scores and figures were calculated from the labeled BBC dataset in the executed Colab notebook. The full experiment uses 1,440 articles; full spaCy grammatical and entity analysis uses 175 stratified articles. The two illustrative integration checks are distinct from independent generalization tests. Code and source citations are available in the linked repository.

References

BBC News Classification, Kaggle: https://www.kaggle.com/competitions/learn-ai-bbc/data

spaCy: https://spacy.io/ | scikit-learn: https://scikit-learn.org/ | NLTK: https://www.nltk.org/